THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET A

Chapter 1: **A Square and a Cube** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 144	$12^2 = 144$
2	**(a)** 169	$13 \times 13 = 169$
3	**(b)** 14	$14 \times 14 = 196$
4	**(a)** 33	$33 \times 33 = 1089$
5	**(c)** 225	Sum of first n odd numbers $= n^2 = 225$
6	**(a)** 28	There are $2n = 28$ numbers between n^2 and $(n+1)^2$
7	**(d)** 3375	$15 \times 15 \times 15 = 3375$
8	**(b)** 15	$15^3 = 3375$
9	**(b)** 1000	$10^3 = 1000$
10	**(d)** 9	$7^2 = 49$ ends in 9

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Yes, $576 = 24 \times 24$, so $\sqrt{576} = 24$.	1+1
12	Yes, $1225 = 35 \times 35$, so $\sqrt{1225} = 35$.	1+1
13	Yes, $2304 = 48 \times 48$, so $\sqrt{2304} = 48$.	1+1
14	The sum of the first n odd numbers is n^2 (1); so the sum is $5^2 = 25$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$3375 = 3^3 \times 5^3$ (2); cube root $= 15$ (1).	2+1
16	$5832 = 2^3 \times 3^6$ (2); cube root $= 18$ (1).	2+1
17	$10648 = 2^3 \times 11^3$ (2); cube root $= 22$ (1).	2+1
	OR: $17576 = 2^3 \times 13^3$ (2); cube root $= 26$ (1).	2+1
18	Factorise 2025 into prime pairs (2); the square root is 45 (1).	2+1
	OR: Factorise 3136 into prime pairs (2); the square root is 56 (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $1+3+5+7+9+11 = 36$ (2). (ii) $25^2 = 625$ (1.5). (iii) $40^2 = 1600$ so 40 (1.5).	2+1½+1½
	OR: (i) $7569 = 3^2 \times 29^2$, root 87 (2.5). (ii) $3^2 \times 29^2$ needs 3×29 for a cube, i.e. 87 (2.5).	2½+2½
20	(i) $12167 = 23^3$, root 23 (2). (ii) $3240 = 2^3 \times 3^4 \times 5$; divide by $3 \times 5 = 15$ to get 216 (3).	2+3
	OR: (i) $24^2 = 576$ m² (1). (ii) tile area 0.25; $576/0.25 = 2304$ tiles (2). (iii) $1.5^3 = 3.375$ m³ $= 3375$ litres (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 1: **A Square and a Cube** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 225	$15^2 = 225$
2	**(a)** 289	$17 \times 17 = 289$
3	**(d)** 18	$18 \times 18 = 324$
4	**(c)** 42	$42 \times 42 = 1764$
5	**(c)** 400	Sum of first n odd numbers $= n^2 = 400$
6	**(d)** 216	$6 \times 6 \times 6 = 216$
7	**(a)** 7	$7^3 = 343$
8	**(d)** 21	$21^3 = 9261$
9	**(a)** 7	Squares end only in 0, 1, 4, 5, 6 or 9
10	**(d)** negative	negative × negative × negative = negative

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The sum of the first n odd numbers is n^2 (1); so the sum is $9^2 = 81$ (1).	1+1
12	The sum of the first n odd numbers is n^2 (1); so the sum is $12^2 = 144$ (1).	1+1
13	A number that is the square of a natural number (1); e.g. 16 and 49 (1).	1+1
14	A number that is the cube of a natural number (1); e.g. 27 and 125 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Factorise 4624 into prime pairs (2); the square root is 68 (1).	2+1
16	Factorise 7396 into prime pairs (2); the square root is 86 (1).	2+1
17	$180 = 2^2 \times 3^2 \times 5$ (1); multiply by 5 (1); 900, $\sqrt{900} = 30$ (1).	1+1+1
	OR: $500 = 2^2 \times 5^3$ (1); divide by 4 (1); quotient $125 = 5^3$ (1).	1+1+1
18	$64 = 8^2$ (1); so the sum of the first 8 odd numbers (1): $1+3+5+7+9+11+13+15$ (1).	1+1+1
	OR: For odd m, triplet $(m, \frac{m^2-1}{2}, \frac{m^2+1}{2})$ (1); $m = 9$ gives 40 and 41 (1); $9^2 + 40^2 = 41^2$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $(5,12,13)$, $(7,24,25)$, $(9,40,41)$ (3). (ii) $5^2 + 12^2 = 25 + 144 = 169 = 13^2$ (2).	3+2
	OR: (i) $4^3 = 64 = 13 + 15 + 17 + 19$ (3). (ii) $125 = 21 + 23 + 25 + 27 + 29$ (2).	3+2
20	(i) Smallest number that is the sum of two cubes in two different ways (1). (ii) $1^3 + 12^3$ and $9^3 + 10^3$ (2). (iii) $1 + 1728 = 1729$; $729 + 1000 = 1729$ (2).	1+2+2
	OR: (i) $\frac{21}{28} = \frac{3}{4}$ (2.5). (ii) $\frac{7}{12}$ (2.5).	2½+2½

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.